KOOPMAN MODEL PREDICTIVE CONTROL FOR COMBINED YAW AND THRUST CONTROL UNDER TURBULENT INFLOW

Antje Dittmer, Bindu Sharan, Joshua H. Merlis, Herbert Werner
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ICS Institute of Control Systems

TUHH Technische Universität Hamburg

DLR

Credit: DLR

Motivation Wind Farm Control

- Wind farms
  - Maximize possible yield per available area, minimize operation and maintenance cost
  - Wake interaction
    - power losses and
    - additional mechanical loads on downstream turbines



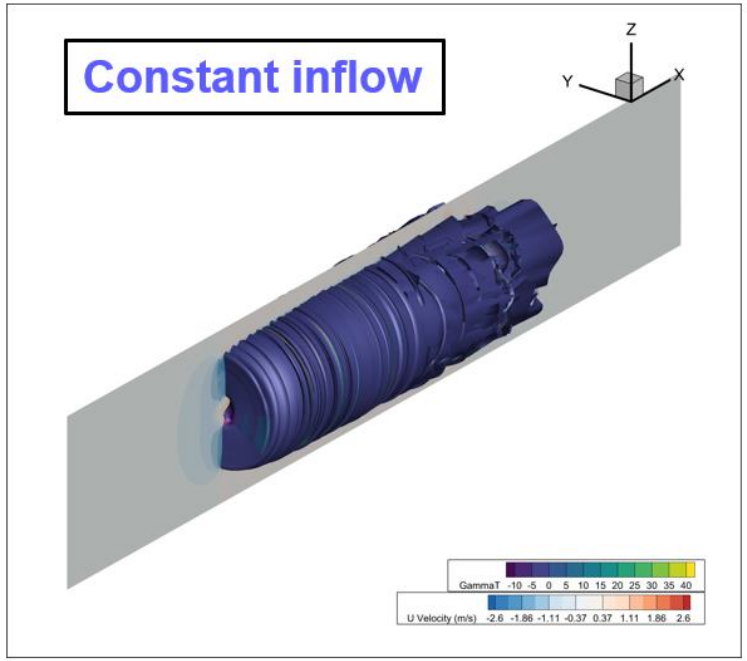
Courtesy: Vattenfall; photographer is Christian Steiness.

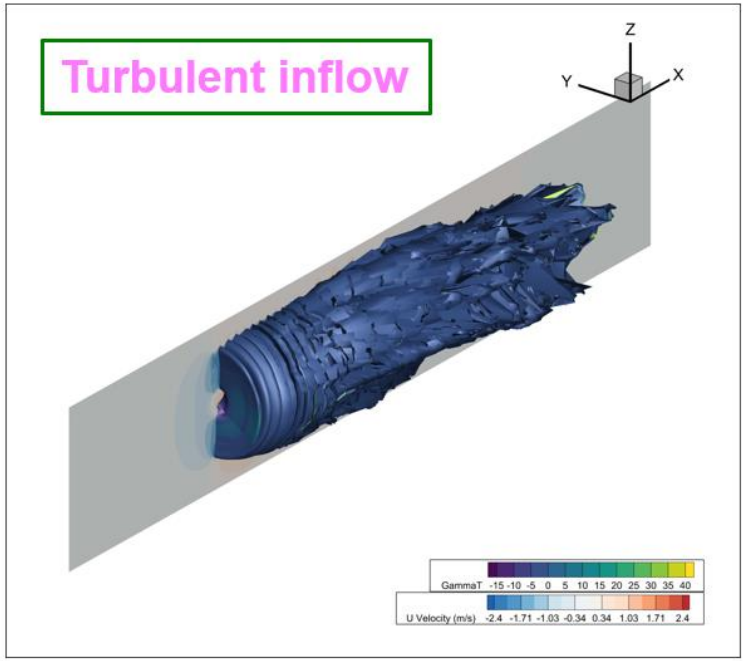
- Wind farm control objectives
  - Minimize error to reference power / maximize generated power
  - Minimize damage equivalent loads (DEL)
  - Minimize changes in control input

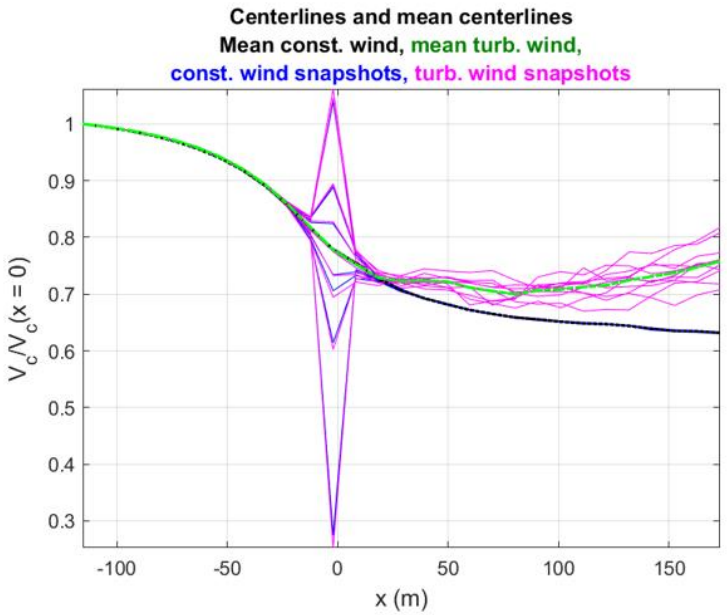
- Control strategies
  - Axial induction control (AIC), e.g. [Pedersen et. al 2020], [van der Hoek et al. 2019]
  - Wake redirection control (WRC), e.g. [Fleming et al. 2020], [Simley et al. 2022]

Influence Turbulence Intensity on Wake Recovery

- WRC effectiveness depends on wake recovery, which depends on turbulence intensity
- Differences in wake recovery visible in simulation, e.g., DLR‘s MAECOsim®







- Mid-fidelity models: Constant wind tends to overpredict influence of wake, high-fidelity models: Turbulent inflow condition show earlier wake recovery
- WRC for power output maximization shown to be still effective in FLORIDyn [Becker et al. 2025]

Koopman Model Predictive Control for Wind Farms

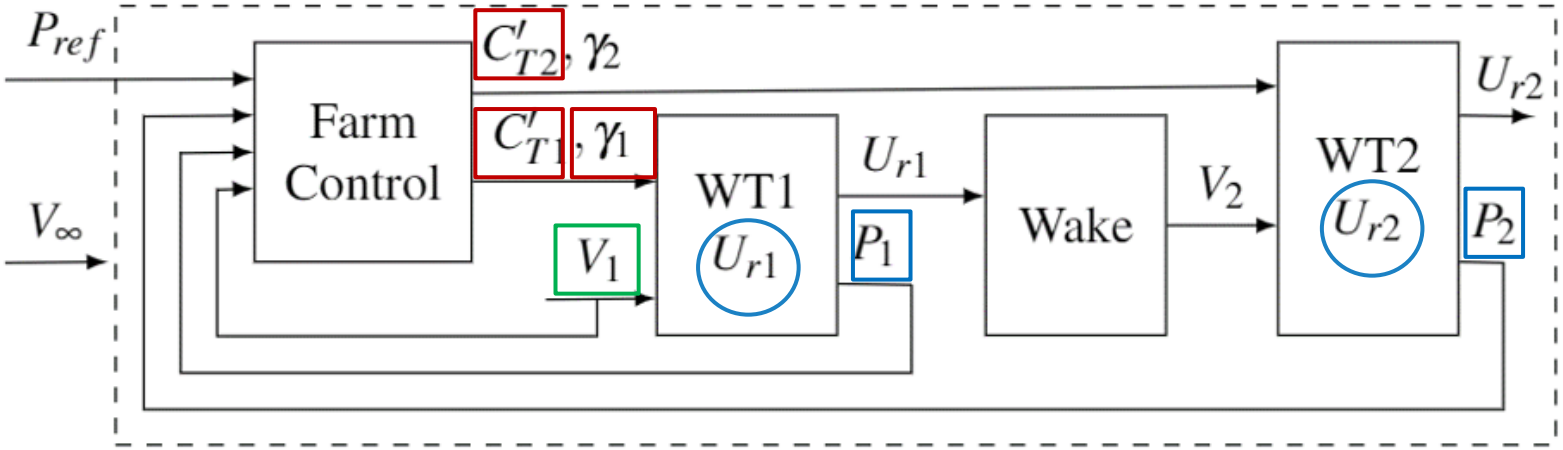
- Koopman model predictive control (MPC) [Cassamo et al. 2021]
  - 100.000 grid points of 3D wind field → flattened 3D structure to vectors, applied Koopman lifting functions
  - Singular value decomposition for model order reduction and dynamic mode decomposition → MPC

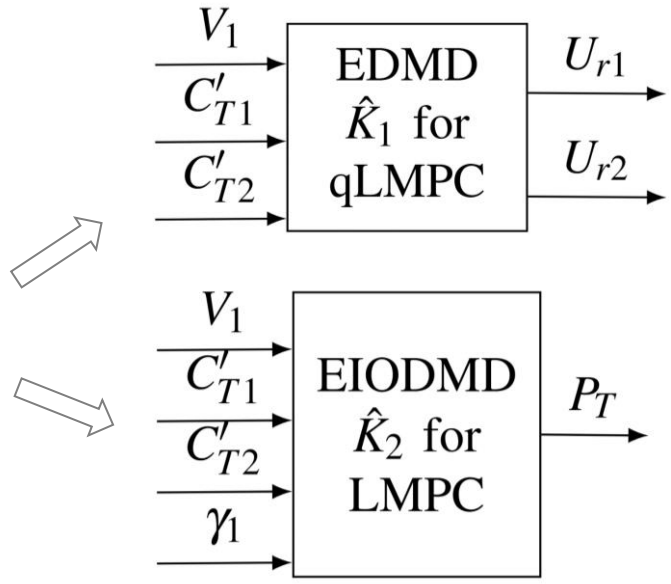
- Previous work: Realtime Koopman MPC in WFSim based on considerably fewer wind measurements
  - AIC MPC: Based on effective wind speeds [Sharan et al. 2022], adaptive MPC based on effective wind speeds and one additional wind measurement [Dittmer et al. 2022]
  - Combined thrust and yaw control (WRC-AIC) [Dittmer et al. 2023]

- Current work: WFSim environment extended with turbulent inflow:
  - Open/Loop: What will be the effect of turbulence on wake recovery and farm power increase in WFSim?
  - Closed loop: influence of turbulence on WRC potential for power reference tracking, tested with three MPC schemes (baseline AIC and two WRC-AIC) from [Dittmer et al. 2023]

WFSim Data Acquisition and Closed-Loop Simulations

- WFSim [Boersma et al. 2016]: 2D Navier-Stokes equations, linearized on a structured staggered grid
- Open-loop data acquisition with turbulent inflow V_∞
  - Inputs: **Thrust controls** C'_{T1} and C'_{T2}, **yaw control** γ_1, **disturbance wind** V_1
  - States: **Average wind** speeds acting on rotor discs U_{r1} and U_{r2}
  - Output: **Total farm power** $P_T = P_1 + P_2$
  - External dynamic mode decomposition (EDMD), external input-output DMD (EIODMD)
- Closed-loop controller tests: Grid power reference P_{ref}

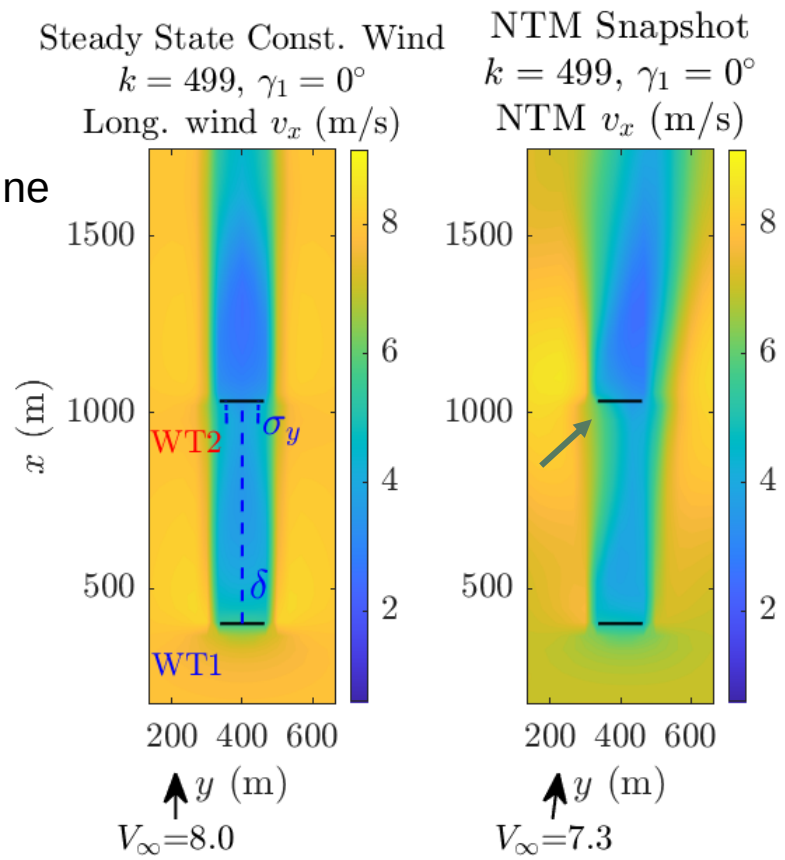


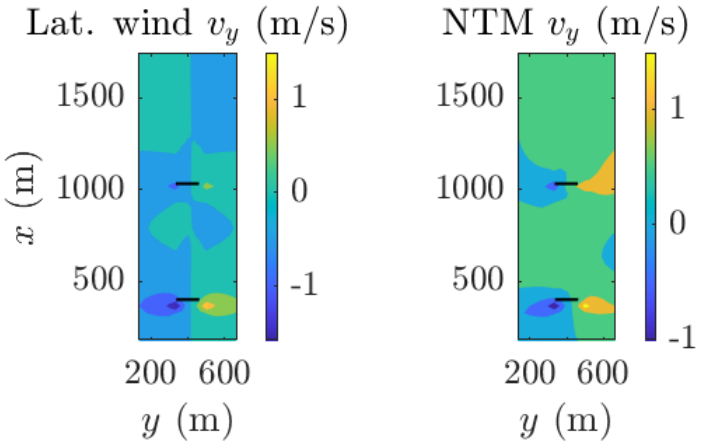


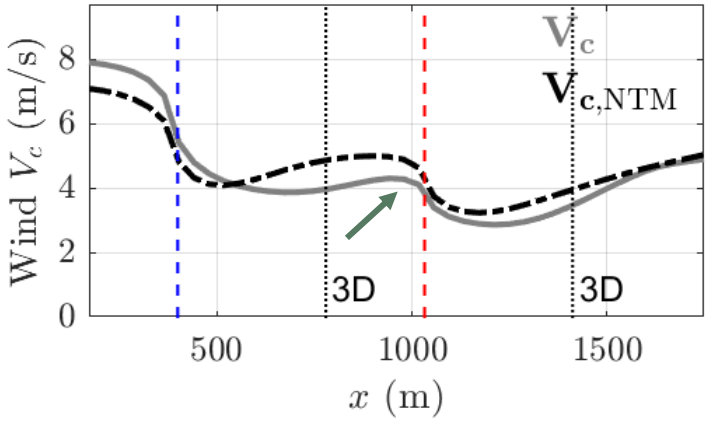
Open Loop WFSim: Constant vs. Turbulent Inflow (γ1 = 0°)

Steady state matches centerline δ and wake extension σ_y of the Gaussian model

NTM snapshot (TI = 10%): more rapidly diverging, meandering wake







$V_{c,NTM}$ is better recovered at the WT2 than V_c under constant wind

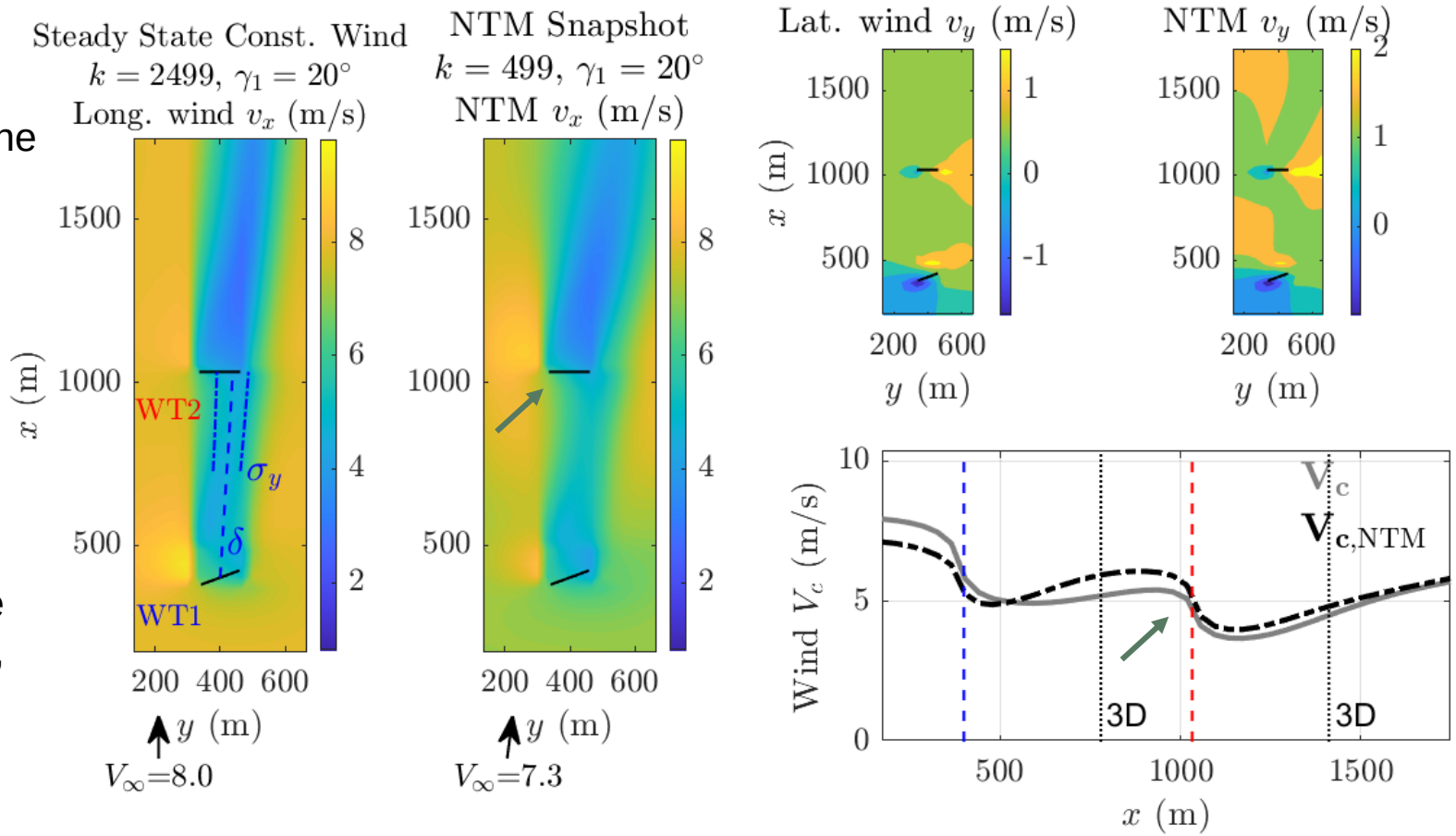
Wind fields and centerline wind amplitudes at turbines WT1 and WT2

Open Loop WFSim: Constant vs. Turbulent Inflow (γ1 = 20°)

Steady state matches centerline δ and wake extension σ_y

Steady state steering: clear wake deflection past WT2

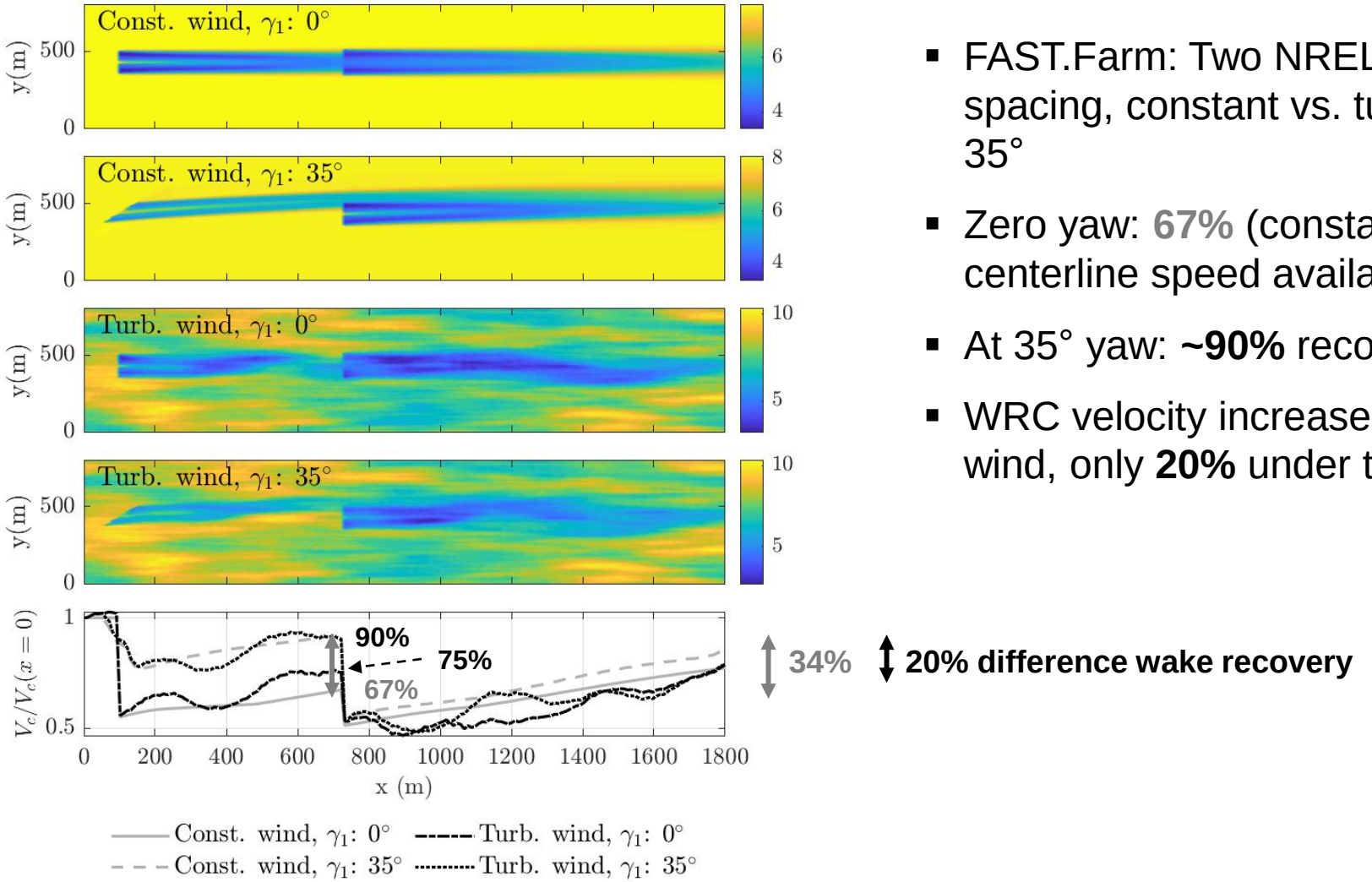
NTM snapshot (TI = 10%): more rapidly diverging, meandering wake → deflection partly washed out



$V_{c,NTM}$ recovers only slightly faster than V_c under constant wind →Faster recovery reduces the relative benefit of active yaw control

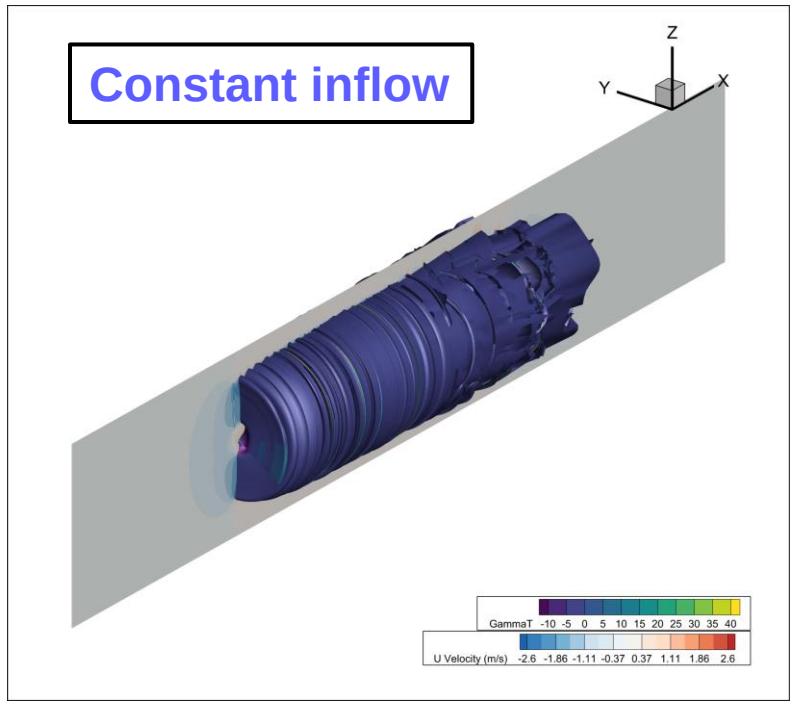
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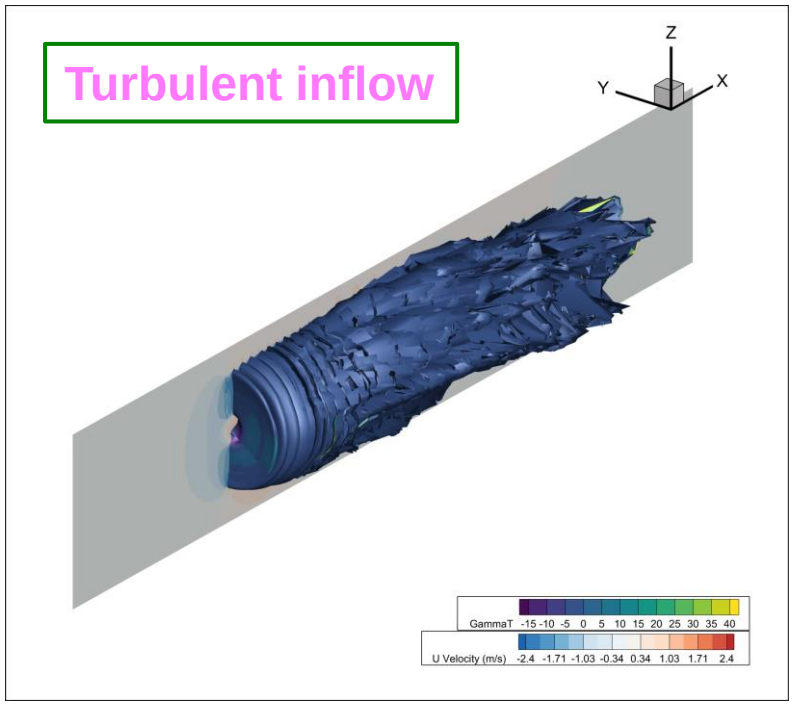
Open Loop Simulation: Verification with FAST.Farm

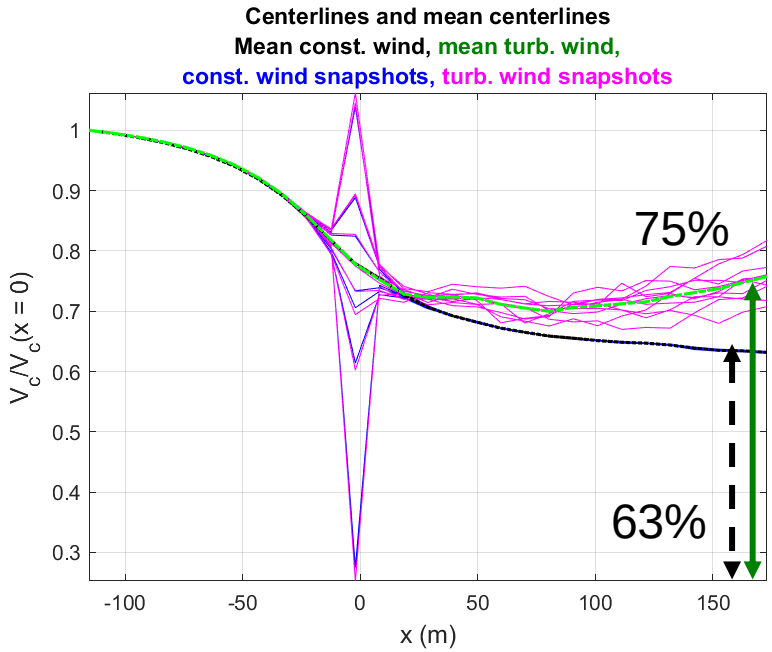


- FAST.Farm: Two NREL 5 MW turbines at 5D spacing, constant vs. turbulent inflow, γ1 = 0° and 35°
- Zero yaw: 67% (constant) vs. **75%** (turbulent) of centerline speed available at downstream turbine
- At 35° yaw: **~90%** recovery at 5D in both cases
- WRC velocity increase at 5D: 34% under constant wind, only **20%** under turbulent wind

Open Loop Simulation: Verification MAECOsim®



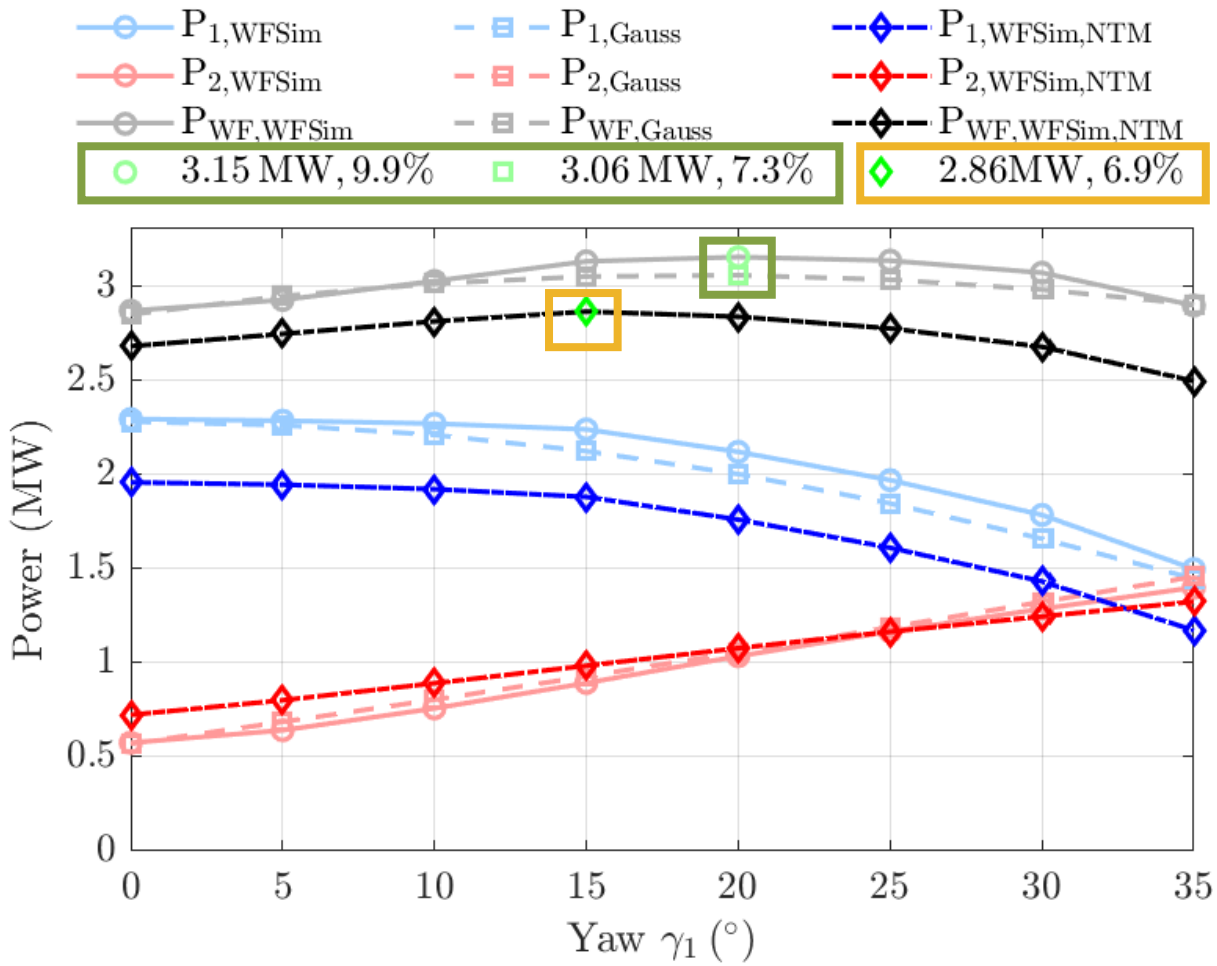




- Turbulence-dependent recovery also confirmed with DLR's MAECOsim® [Michels and Seelemeyer 2024]

→ FAST.Farm and MAECOsim® show the same qualitative trends as WFSim: Turbulence increases wake recovery

Open Loop Sim.: Power Yield with Turbulent Inflow

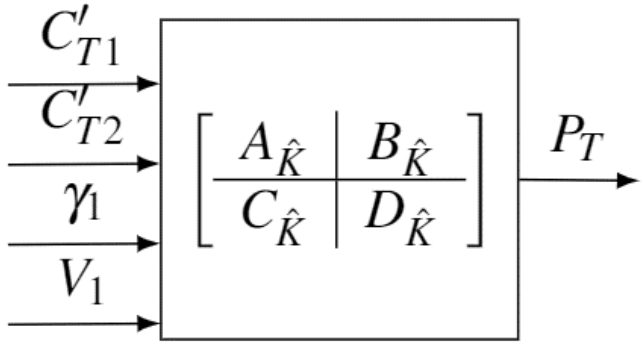


- Yaw sweep γ1 = 0° … 35°, mean power for t ∈ [300, 1000] s
- Faster natural recovery shifts the optimum from 20° to 15°: downstream gain no longer compensates a larger upstream sacrifice

→WRC remains beneficial, but gains shrink from ~10% to ~7%

System Identification with Koopman

- Use extended input-output dynamic mode decomposition
- Data:
  - $X = [x_1 \quad x_2 \quad \cdots \quad x_{N-1}]: \mathbb{R}^{n_x \times n_{N-1}}$, $X_+ = [x_2 \quad x_3 \quad \cdots \quad x_N]: \mathbb{R}^{n_x \times n_{N-1}}$
  - $U = [u_1 \quad u_2 \quad \cdots \quad u_{N-1}]: \mathbb{R}^{n_u \times n_{N-1}}$
  - $Y = [y_1 \quad y_2 \quad \cdots \quad y_{N-1}]: \mathbb{R}^{n_y \times n_{N-1}}$



- Koopman lifting functions: State x_k and its non-linear combinations $g(x_k)$:

$$\Phi_k^\top = \left[x_k^\top \; g(x_k)^\top\right]: \mathbb{R}^{n_x} \times \mathbb{R}^m \to \mathbb{R}^{n_\varphi}$$

Lift and append data:
$L_u^\top = [\Phi(X)^\top \; U^\top]$
$L_+^\top = [\Phi(X_+)^\top \; Y^\top]$

→

Formulate:
$\min_{\hat{K}} ||L_+ - \hat{K} L_u||_F$

→

Solve:
$\hat{K} = L_+ L_u^\dagger = \begin{bmatrix} A_{\hat{K}} & B_{\hat{K}} \\ C_{\hat{K}} & D_{\hat{K}} \end{bmatrix}$

Koopman MPC for Wind Farms

- Objectives for wind farm MPC: Minimize **error** and **change in turbine controls**

$$\min_u J(u) = e^T Q e + \Delta u^T R \Delta u$$

$$s.t.$$

$$\begin{bmatrix} \Phi_{k+1} \\ P_{T,k} \end{bmatrix} = \begin{bmatrix} A_{\hat{K}} & B_{\hat{K}} \\ C_{\hat{K}} & D_{\hat{K}} \end{bmatrix} \begin{bmatrix} \Phi_k \\ u_k \end{bmatrix}, \quad k \in \{1, 2, \ldots, n_h\}$$

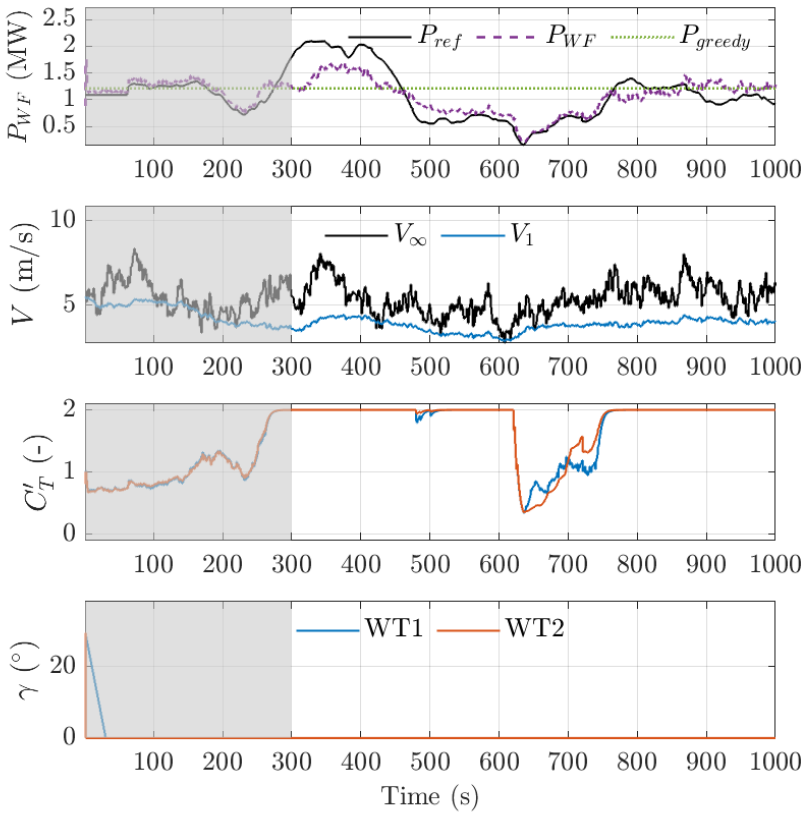
$$e = P_{ref} - P_T = [e_1, e_2, \ldots, e_{n_h}]^T \in \mathbb{R}^{n_h}$$

$$\Delta u = \left[u_1^T - c^T, u_2^T - u_1^T, \ldots, u_{n_h}^T - u_{n_h-1}^T\right]^T \in \mathbb{R}^{n_h n_u}$$

- Use **Koopman wind farm models** to predict wind farm states over a **finite time horizon**
- Solve **optimization problem** to obtain **optimal control inputs** for each turbine under input constraints
- Set the next control inputs and update predictions and optimization problem at each time step

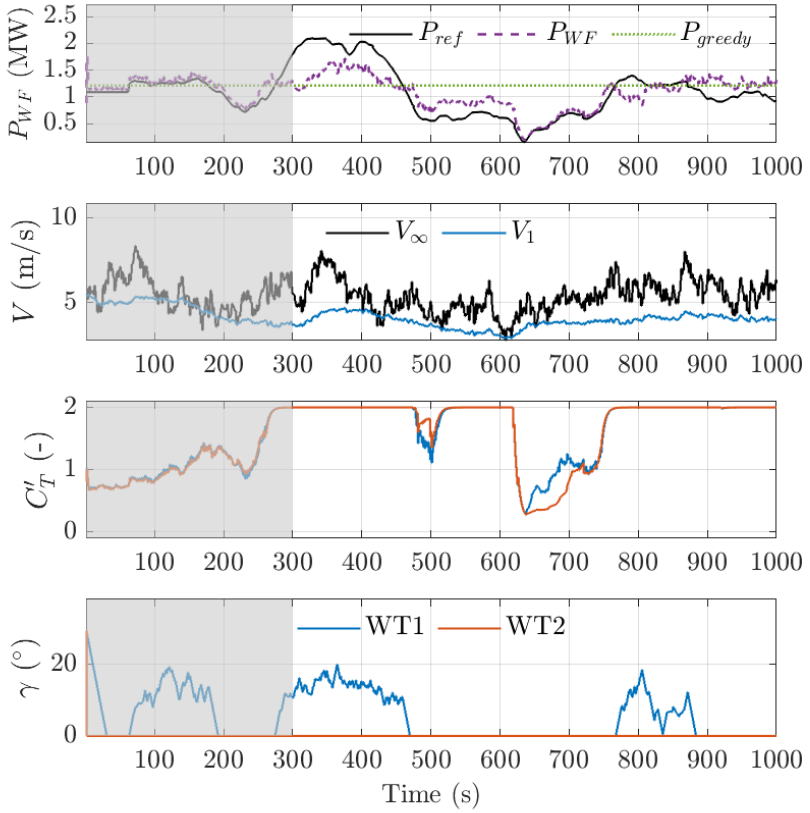
Closed Loop Simulation (1): EDMD Koopman qLMPC

Baseline thrust control (AIC); TE: 226 kW



- Reference tracking with thrust control only

qLMPC with EDMD/static yaw+AIC; TE: 253 kW

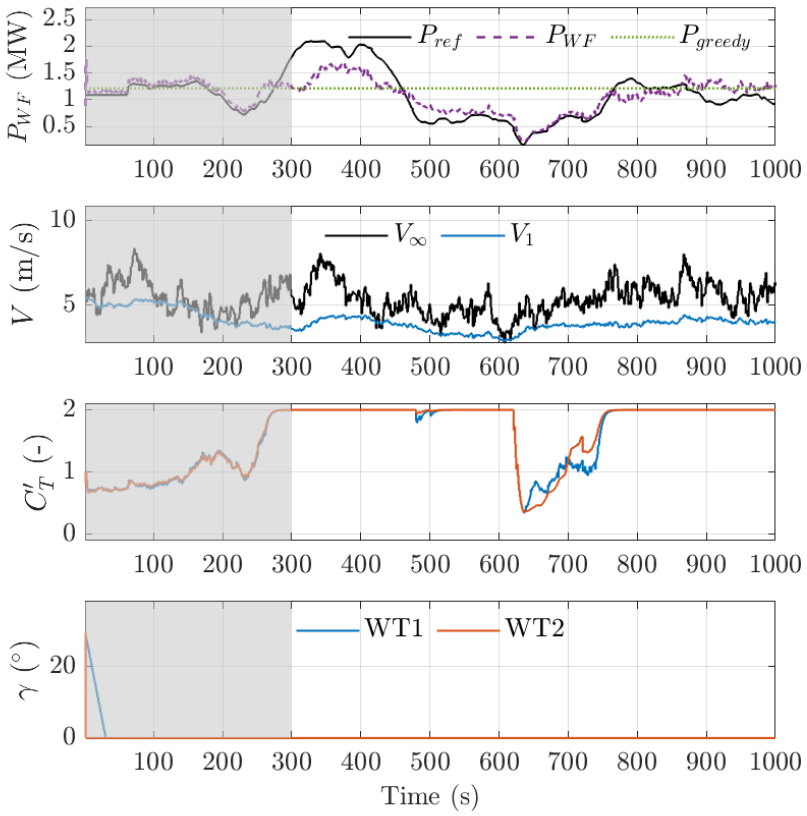


- WT1 and WT2 thrust control at yaw angle γ^*

→ Static yaw increases (instead of decreases) tracking error

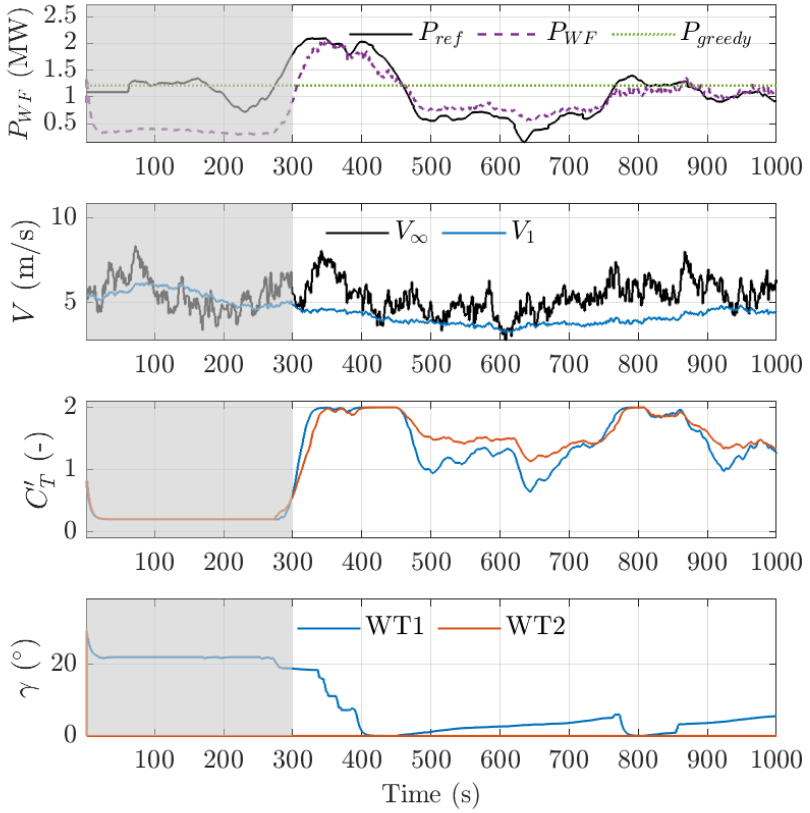
Closed Loop Simulation (2): EIODMD Koopman LMPC

Baseline thrust control (AIC); TE: 226 kW



- Reference tracking with thrust control only

LMPC w. EIODMD (WRC-AIC); TE = 158 kW



- Reference tracking with combination of yaw and thrust control

Summary

- Open loop:
  - WFSim extended with Kaimal NTM turbulent inflow (IEC 61400-1)
  - Turbulence accelerates wake recovery → confirmed with FAST.Farm and MAECOsim®
  - WRC potential reduced from 9.9% to 6.9%; optimal yaw shifts from 20° to 15°

- Closed loop: joint Koopman WRC-AIC outperforms baseline AIC and hybrid scheme
  - Tracking error reduced by 30% (158 kW vs. 226 kW) with small yaw excursions

- Outlook
  - Tracking errors increased vs. steady-state studies → room for model improvements
  - Additional observables and neural networks to better handle turbulent wind

Code: github.com/antjedittmer/2026-code-Koopman-MPC-under-Turbulent-Inflow

Thank you very much for your attention!